

Problem Set 4 — Structural VARs: Short-Run and Long-Run Identification, and VECMs

Important

We **do not collect** this problem set, and you **do not need to do all exercises**. We will cover the solutions in class and provide written solutions afterwards for your study. The goal is to give you more material to prepare for the exam.

Problem 1 (Hybrid Identification: Short and Long-Run Restrictions). This problem follows the identification strategy in Galí (1999), where technology shocks are identified by imposing that only technology has a permanent effect on labor productivity, and combines it with the short-run timing logic in Christiano et al. (2005), where a monetary policy shock is identified using contemporaneous exclusion restrictions. The treatment of the reduced-form VAR and of the structural impact matrix follows standard references such as Lütkepohl (2005). Let $y_t = (\Delta a_t, n_t, \pi_t, i_t)'$ where Δa_t denotes labor productivity growth (or a TFP-growth proxy), n_t hours (or an activity measure), π_t inflation, and i_t a short-term policy rate. Assume y_t follows a stable reduced-form VAR(p),

$$y_t = \sum_{j=1}^p A_j y_{t-j} + u_t, \quad \mathbb{E}[u_t u_t'] = \Sigma_u. \quad (1)$$

Consider a structural representation

$$u_t = B_0^{-1} w_t, \quad \mathbb{E}[w_t w_t'] = I_4, \quad (2)$$

where $w_t = (\varepsilon_t^a, \varepsilon_t^d, \varepsilon_t^\pi, \varepsilon_t^m)'$ denote, respectively, a technology shock, a non-technology real (demand-type) shock, a cost-push/inflation shock, and a monetary policy shock. The structural moving-average representation of this model is $y_t = \sum_{h \geq 0} \Theta_h w_{t-h}$, and define the *long-run multiplier* $\Theta(1) = \sum_{h \geq 0} \Theta_h$.

- Write the restriction used in Galí (1999) in a form that is operational in this VAR. Your restriction must express that **only** ε_t^a has a long-run effect on labor productivity in levels. Be explicit about how this is written when the VAR includes Δa_t (so that the long-run effect on the *level* of productivity involves the cumulative effect of Δa_t).
- Write the short-run restriction in the spirit of Christiano et al. (2005): productivity growth does not respond contemporaneously to a monetary policy shock, i.e.

$$\left. \frac{\partial(\Delta a_t)}{\partial \varepsilon_t^m} \right|_t = 0.$$

State the corresponding zero restriction on the appropriate entry of B_0^{-1} (the impact matrix).

- (c) Determine whether the model is exactly identified by your restrictions. First perform a **restriction count** relative to the degrees of freedom in B_0^{-1} under the normalization $\mathbb{E}[w_t w_t'] = I_4$. Then explain why the count alone is not sufficient and state, in words and with the relevant mapping, the **rank condition** you would check for local identification.
- (d) Provide a **step-by-step computational procedure** that takes $(\hat{A}_1, \dots, \hat{A}_p, \hat{\Sigma}_u)$ as inputs and returns an estimate $\hat{B}_0^{-1}$ under the joint restrictions. Your description must specify the reduced-form long-run covariance object constructed from $A(1)$ and Σ_u , and how the short-run restriction is enforced jointly with the long-run restriction.
- (e) Discuss, without estimating, what pattern of responses of n_t and π_t to the identified technology shock ε_t^a would be consistent with Galí's empirical finding that hours may *fall* after a positive technology shock, and reconcile this with the imposed long-run effect on productivity.

Problem 2 (Permanent and Transitory Shocks in a Cointegrated System: SVECM). This problem mirrors the permanent–transitory identification logic in [Shapiro & Watson \(1988\)](#) and [King et al. \(1991\)](#), where a cointegrated system is used to separate permanent from transitory shocks using long-run restrictions. The structural VECM framework is as in [Lütkepohl \(2005\)](#). Let $x_t = (y_t, c_t, i_t)'$ denote log output, log consumption, and log investment, and suppose x_t is $I(1)$ with cointegration rank $r = 1$. Consider the reduced-form VECM

$$\Delta x_t = \alpha \beta' x_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta x_{t-j} + u_t, \quad \mathbb{E}[u_t u_t'] = \Sigma_u, \quad (3)$$

where α is 3×1 and β is 3×1 , so $\alpha \beta'$ is the usual loading–cointegrating matrix. Let the structural shocks satisfy $u_t = B_0^{-1} w_t$ with $\mathbb{E}[w_t w_t'] = I_3$, where $w_t = (\varepsilon_t^P, \varepsilon_t^T, \varepsilon_t^M)'$ are a permanent shock, a transitory shock, and a monetary/financial shock. Let Υ denote the *structural long-run impact matrix on levels*: the cumulative effect of w_t on x_t as the horizon goes to infinity.

- (a) Using the fact that the cointegration rank is $r = 1$, determine the number of common stochastic trends in the system and state the **maximum number** of linearly independent permanent shocks that can exist. Explain briefly why this restriction arises from cointegration rather than from identifying assumptions.
- (b) Impose the Shapiro–Watson / KPSW-style restriction that the **transitory** shock has no long-run effect on any of the levels y_t , c_t , and i_t . Express this as zero restrictions on the appropriate column of Υ , and state how many independent long-run restrictions this gives you.
- (c) Impose an additional long-run restriction consistent with neutrality: the monetary/financial shock has no long-run effect on either y_t or on $y_t - c_t$. Choose one of these two formulations, justify your choice, and write the restriction as an additional zero restriction on Υ .

- (d) Add one **short-run** restriction: the monetary/financial shock does not move output growth contemporaneously in the SVECM,

$$\left. \frac{\partial(\Delta y_t)}{\partial \varepsilon_t^M} \right|_t = 0.$$

Translate this into a zero restriction on B_0^{-1} .

- (e) Analyze **identification** under the restrictions in parts (b)–(d). Include a restriction count for the free parameters in B_0^{-1} under $\mathbb{E}[w_t w_t'] = I_3$, and discuss one way these restrictions can fail to be feasible because of the rank structure of the long-run impact matrix implied by cointegration.
- (f) Explain why estimating a **VAR in first differences** $\Delta x_t = \sum_j \tilde{A}_j \Delta x_{t-j} + \tilde{u}_t$ is *not* a valid alternative representation for this cointegrated system. (*Hint*: what does the Beveridge–Nelson decomposition imply about the Wold representation of Δx_t when $\psi(1)$ is singular?) Given this, compare implementing the long-run restrictions in the **SVECM directly** versus in an unrestricted **VAR in levels**. State one advantage and one disadvantage of each approach, and give a precise failure mode when the cointegration rank is misspecified in the SVECM.

Problem 3 (Recursive identification under misspecification: the CFP critique). This problem is based on [Carlstrom et al. \(2009\)](#): if the true data-generating process is forward-looking and the econometrician imposes recursive short-run identification (e.g. Cholesky) that is inconsistent with the true contemporaneous transmission, then the identified “monetary policy shock” can be a mixture of true structural shocks and impulse responses can be severely distorted. Let the true DGP for $z_t = (y_t, \pi_t, i_t)'$ admit a structural MA representation

$$z_t = \sum_{h=0}^{\infty} \Psi_h \varepsilon_{t-h}, \quad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I_3, \quad (4)$$

where $\varepsilon_t = (\varepsilon_t^a, \varepsilon_t^c, \varepsilon_t^m)'$ denote technology, cost-push/demand, and monetary policy shocks. An econometrician estimates a reduced-form VAR(p)

$$z_t = \sum_{j=1}^p A_j z_{t-j} + u_t, \quad \mathbb{E}[u_t u_t'] = \Sigma_u, \quad (5)$$

and identifies structural shocks by imposing a **Cholesky scheme** with ordering (y, π, i) , so that $u_t = B_0^{-1} w_t$ with B_0^{-1} lower triangular and $\mathbb{E}[w_t w_t'] = I_3$, where w_t is the vector of Cholesky-identified structural shocks (the econometrician’s recovered “structural” residuals). The econometrician interprets the third entry w_t^m as the monetary policy shock.

- (a) Write the lower-triangular structure of B_0^{-1} implied by the Cholesky ordering (y, π, i) . Interpret each zero as a contemporaneous exclusion restriction: which variables are allowed to respond contemporaneously to which structural shocks under this scheme?

- (b) Prove that, without additional identifying restrictions, the mapping from reduced-form shocks u_t to structural shocks is not unique because Σ_u only pins down B_0^{-1} up to an orthogonal rotation. Then explain how the Cholesky scheme selects a particular rotation and why this is an economic assumption rather than a statistical necessity.
- (c) Assume that in the **true** DGP, monetary policy affects output contemporaneously (e.g. through expectations or financial channels), so that the impact response satisfies $(\Psi_0)_{y,m} \neq 0$, while the Cholesky identification with ordering (y, π, i) imposes $(B_0^{-1})_{y,m} = 0$. Show that, under invertibility of the relevant linear mappings, the identified “monetary shock” w_t^m must be a nontrivial linear combination of the true shocks $(\varepsilon_t^a, \varepsilon_t^c, \varepsilon_t^m)$.
- (d) Give a **sufficient condition** under which the impulse response of y_t to the identified Cholesky monetary shock w_t^m can have the *opposite sign* of the impulse response of y_t to the true monetary policy shock ε_t^m . Your condition should be stated as a restriction on the contemporaneous impact matrix and at least one additional feature of the dynamics.
- (e) Propose **two robustness strategies** that directly address the Carlstrom–Fuerst–Paustian critique. For each, state what identifying assumption replaces the recursive contemporaneous exclusions and what object is identified as a result.

References

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